

of benchmark was warranted. However, what ultimately mattered for GM was meeting its pension fund liability. Herscovici, a global equities portfolio manager, believed that “the low volatility strategy has two main factors consistent with GM’s direction to reduce surplus volatility: lower risk and a higher correlation to pension liabilities.”

Most of all, Scott and Herscovici wondered what explained this anomaly. If it offered such a great opportunity, why weren’t a lot of other investors doing it? And would it persist going forward?

2. Active Management

Alpha is often interpreted as a measure of skill. That is debatable. It is first and foremost a statement about a benchmark.

2.1. DEFINITION OF ALPHA

Alpha is the average return in excess of a benchmark. Thus, the concept of alpha requires first defining a benchmark against which alpha can be measured.

We define the excess return, r_t^{ex} , as the return of an asset or strategy in excess of a benchmark as

$$r_t^{ex} = r_t - r_t^{bmk}, \quad (10.1)$$

for r_t the return of an asset or strategy and r_t^{bmk} the benchmark return. Sometimes we refer to the excess returns, r_t^{ex} , as *active returns*. This terminology assumes that the benchmark is *passive* and can be produced without any particular investment knowledge or even human intervention. Common passive benchmarks are market-weighted portfolios, like the S&P 500 or the Russell 1000, which investors can track by buying low-cost index funds.

We compute the alpha by taking the average excess return in equation (10.1):

$$\alpha = \frac{1}{T} \sum_{t=1}^T r_t^{ex}, \quad (10.2)$$

where there are T observations in the sample.

The other two terms that we need to define are *tracking error* and the *information ratio*. Tracking error is the standard deviation of the excess return; it measures how disperse the manager’s returns are relative to the benchmark:

$$\text{Tracking Error} = \bar{\sigma} = \text{stdev}(r_t^{ex}). \quad (10.3)$$

Tracking error constraints are imposed to ensure a manager does not stray too far from the benchmark (see chapter 15 on agency issues in delegated portfolio management). The larger the tracking error, the more freedom the manager